

Introduction to R, RStudio, and plots

In this document I will introduce the R software, along with its programming language. I will use this software throughout the course, and you will need to use the software for certain purposes on homework. Computers are powerful tools for statisticians and it will be helpful for you to be familiar with the basics of programming. R can be used to do simple calculations, create plots and figures, and run statistical analyses. If you have any questions or are curious about other aspects of R, there is a lot of material online that is worth reading.

Additional reading: http://cran.r-project.org/doc/contrib/Paradis-rdebuts_en.pdf
(especially pages 1-2, 5, 9, 15, 25-28, 32-33, 40-43)

Downloading and installing R

We will be using RStudio, which is an open-source user interface to the R software. To get it on your personal computer there is a two-step process.

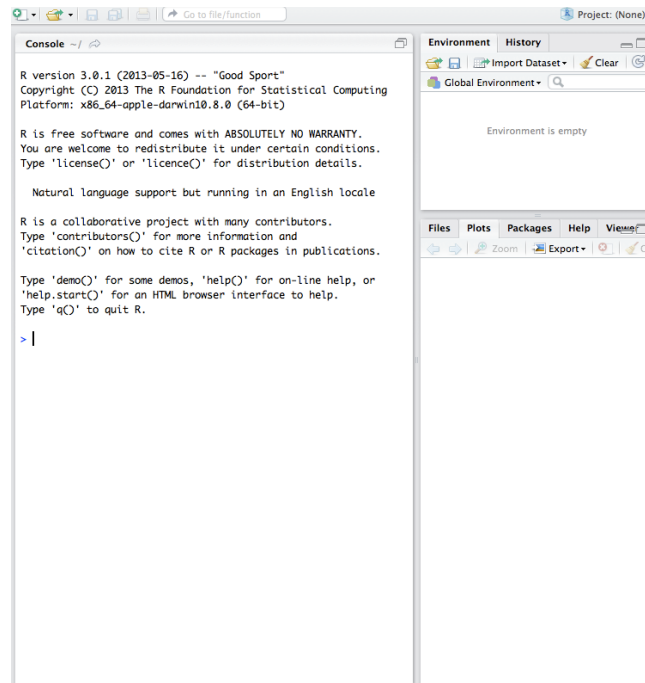
- 1) Install R on your personal computer, following the instructions at <http://cran.r-project.org>
- 2) Install RStudio on your personal computer (<http://www.rstudio.com>)

Note: You have to install R first then install RStudio. This is very important! The instructions are pretty clear on the websites, but if you need help or run into technical difficulties please contact me.

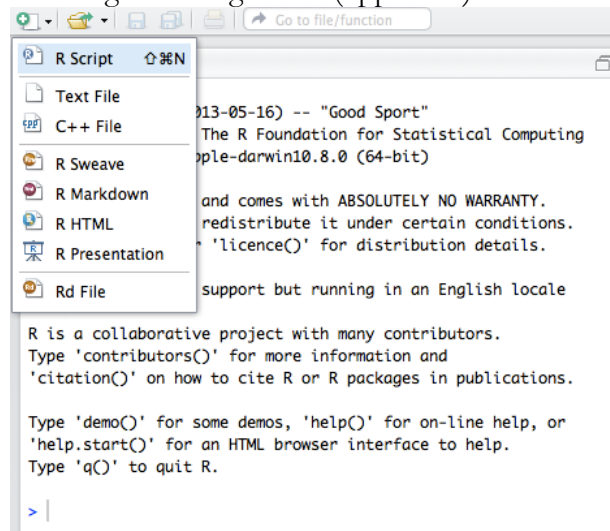
Opening a script

Once you open up RStudio, you'll come to a page that looks like this:

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Click on the white sheet with the green + sign on it (upper left) to create an R-script file.



A new window should appear in the upper left of RStudio. This is where you will type your code. To “run” a line of code, you can highlight the code you wish to run then push Control+Enter. The code you run along with the output will then appear in the “Console” window (on the lower left). **To open an existing script file**, you can click the file icon next to the “Create script” icon in the upper left. You can have multiple script files open at a time, they will appear as tabs above your first script window.

If you haven’t already, download and open the R-script on Canvas titled “Week01-assignment-script.R”.

Commenting in R

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Often in programming languages, you can provide comments within code that explains what certain code does or if you just want to leave notes for yourself. In R, to start a comment it is `#` and everything on the same line immediately to the right of the `#` is commented out.

```
> # Example of a comment. This line does NOT get run by the program  
  
> "This code will run" # but this code won't
```

In the R Script, comments are colored differently from the code. Remember, the `#` starts a comment ONLY on the same line.

Storing scalars and vectors

You can name numeric or character variables any way you want. Spaces are not allowed in naming of variables. It is recommended to only use letters (uppercase or lowercase) and underscores when naming a variable. *R is case sensitive*, which means capital letters are different from lower case letters. Please also use the assignment operator (`<-`) instead of an equals sign. This is standard programming practice.

```
> x <- 1:9  
> x  
[1] 1 2 3 4 5 6 7 8 9  
  
> x <- 6  
> x  
[1] 6  
  
> x + 10  
[1] 16  
  
> x <- x + 10  
> x  
[1] 16
```

Common Functions

Common functions: `mean()`, `sd()`, `min()`, `max()`, `sqrt()`, `exp()`, `log()`, `sum()`
Mean, standard deviation, minimum, maximum, square root, exponential function, logarithm, summation, respectively. There are many more functions available in R but we will learn the necessary functions as we progress through the course.

Help function: The help function is very useful in R. If you are not sure what a function does, type a question mark directly followed by the name of the function. The help page for that function will be brought up in the bottom right of RStudio. Additionally, there are lots of help pages and documentations on the internet for R and its functions.

```
> ?max
```

Reading in external data

There are many different ways of reading in data into R, but we will only use comma-delimited data files (CSV), so the code below will work for all data sets you will be given.

First, make sure you have downloaded the relevant data from Canvas (week01-assignment-data-titanic.csv). For comma-separated data (CSV), you can read it into R using the `read.csv` function, and point to where it is using the `file.choose` function.

```
mydata <- read.csv(file=file.choose())
```

Save your work as a script

Your R scripts can be saved as plain text files with the “.R” extension. To make things easier, I recommend you space out characters and numbers to make your code more readable. Comment often.

Plots in R

Go through the rest of the lab script line by line, and along the way try to figure out what each line of code is doing.

Exercises

Complete the Canvas quiz for Week 01, which goes through the exercises at the end of the R script.